Milestone 2: Scientific Contextualization and Tradeoff Analysis

TTP - A Time-Triggered Protocol

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1. Related Approaches and Neighbouring Paradigms

1.1 Event-Triggered Protocols

CAN uses priority-based - when the bus is free, any node can transmit; conflicts are resolved by bitwise arbitration on the message identifier [2]. This means low-priority frames can be delayed indefinitely if higher-priority traffic keeps arriving. [2, 4].

Table 1: High-level comparison of event- and time-triggered protocols. [3]

Protocol	**Trigger**	**Bus Access**	**Clock Sync**	**Fault Tolerance**	**SAE Class**
CAN	Event	CSMA	Not provided	None built-in	B/C
J1850	Event	CSMA	Not provided	None built-in	B
TTP	Time	TDMA (fixed)	Distributed, μs range	Active (FTU pairs)	C

Key difference vs TTP: ET protocols decide bus access at runtime $\Rightarrow$ non-deterministic worst-case latency. TTP fixes the schedule at design time $\Rightarrow$ deterministic by construction [3] [1] .

1.2 Time-Triggered Protocols

Only TT communication systems are currently considered for X-by-Wire safety-critical applications, because they provide deterministic access to the medium and bounded response times [3]. TTP was designed specifically for SAE Class C — safety-critical, high-speed control in power train and chassis domains [1]. Its integrated services (clock synchronisation, membership, fault tolerance) are precisely the features required at Class C, where failure can have life-critical consequences. Class A and B networks such as LIN and CAN lack these services entirely [3], making them unsuitable for X-by-Wire functions.

SAE Classification [3]:

- **Class A** (< 10 kbps): LIN, TTP/A — body/sensor networks
- **Class B** (10–125 kbps): J1850, low-speed CAN — data sharing between ECUs
- **Class C** (125 kbps–1 Mbps): high-speed CAN, TTP/C, FlexRay — safety-critical, power train, chassis

2. Tradeoff Analysis

2.1 Static Schedule vs. Dynamic / Hybrid Scheduling

TTP uses a purely static TDMA schedule - one message per node per TDMA cycle, fixed at design time [1]. FlexRay improves on this by adding a dynamic segment after the static TDMA segment, allowing event-triggered messages to also be transmitted [4].

Table 2: Scheduling strategy comparison. [1] [4] [3]

Aspect	**TTP (static only)**	**FlexRay (static + dynamic)**	**CAN (dynamic only)**
Worst-case latency	Provably bounded ($\approx$1.92 ms)	Bounded for static segment	Unbounded under load
Aperiodic messages	Poor — wastes slots	Handled in dynamic segment	Efficient
Verification	Fully checkable at design time	Partially checkable	Hard to verify
Flexibility	Only 1 msg/node/cycle	Multiple slots per node	Fully flexible
Runtime overhead	Zero	Low	Medium

Key insight from [3]: A first drawback of purely TT protocols is the inefficiency for aperiodic messages; a second drawback is the lack of flexibility — adding a new transmitting node forces a schedule update on all other nodes. FlexRay's hybrid design directly addresses both weaknesses.

2.2 Fail-Silent vs. Byzantine Fault Model

- TTP assumes nodes either output correct data or nothing (fail-silent).
- Real hardware can fail in non-silent ways — EMI, stuck-at faults, partial transmissions.
- CAN has no babbling-idiot avoidance; TTP and FlexRay both require a bus guardian [4].

Tradeoff: TTP trades generality for simplicity; acceptable for automotive where fail-silence can be enforced via hardware watchdogs.

2.3 Bandwidth Efficiency vs. Worst-Case Latency Guarantee

- TTP allocates bandwidth statically $\Rightarrow$ guaranteed slots even when idle $\Rightarrow$ wasted bandwidth at low load.
- CAN uses bandwidth only when needed $\Rightarrow$ efficient at low load but degrades unpredictably at high load.
- FlexRay at 10 Mbps per channel (20 Mbps total dual channel) [5] is comparable to TTP/C (5–25 Mbps [5]), whereas the original 1993 TTP operated at 250 kbit/s [1] [4].

Table 3: Efficiency and latency comparison.

Metric	**TTP**	**CAN**	**FlexRay**
Data rate	250 kbit/s	1 Mbps	10 Mbps/ch
Data efficiency	53.3%	25.3%	—
Worst-case latency	$\approx$1.92 ms (bounded)	Unbounded under load	Bounded (static segment)

3. Embedded Systems Constraints

Memory Each node stores the full TDMA schedule for every mode ⇒ grows with modes and nodes.

Energy Fixed-schedule transmission allows precise sleep/wake cycles between slots ⇒ suitable for low-power nodes.

Determinism
TT architecture gives full temporal encapsulation - each node is testable in isolation; a local change has no global timing effect [3].

Deployment
Schedule must be regenerated and re-flashed if the topology changes ⇒ high maintenance cost.

Clock hardware
Requires precise quartz crystals; synchronisation in μs range [4]; cheap ceramic resonators insufficient at high speed [2].

References

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[2] Robert Bosch GmbH, *CAN Specification Version 2.0*, Stuttgart, Germany, 1991. Available: `http://esd.cs.ucr.edu/webres/can20.pdf`

[3] N. Navet and F. Simonot-Lion, "A Review of Embedded Automotive Protocols," in *Automotive Embedded Systems Handbook*, N. Navet and F. Simonot-Lion, Eds. Boca Raton, FL: CRC Press, 2009, ch. 4, pp. 4-1–4-27, ISBN 978-0-8493-8026-6.

[4] R. Makowitz and C. Temple, "FlexRay — A communication network for automotive control systems," in *Proc. IEEE Int. Workshop on Factory Communication Systems (WFCS)*, Torino, Italy, 2006, doi: 10.1109/WFCS.2006.1704153.